https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	M is the independent variable and *v* is the dependent variable, or vary *M* and measure *v*	**1**
	keep *x*/compression of spring constant	**1**
	Methods of data collection	
	labelled diagram including horizontal spring in line with vehicle attached to wall / retort stand	**1**
	use a ruler / calliper to determine compression of spring	**1**
	use of stopwatch / use of light gate connected to a timer / motion sensor correctly positioned	**1**
	use of balance to measure mass of vehicle *M*	**1**
	Method of Analysis	
	plots a graph of $1/v^2$ against *M* [Do not allow lg-lg graphs]	**1**
	relationship valid if a straight line produced	**1**
	$k = \frac{1}{gradient \times x^2}$ *or* $k = \frac{b}{y - intercept \times x^2}$	**1**

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Question	Answer	Marks
	Additional detail including safety considerations	**Max 6**
	D1 use safety screen; use goggles to avoid ball / spring hitting eye	
	D2 add masses to the vehicle to change M	
	D3 repeat experiment for each M and average v	
	D4 use of ruler to measure an appropriate distance for the time taken in stopwatch / light gate methods	
	D5 method to determine speed of vehicle, e.g. time vehicle over a measured distance and use speed = distance/time	
	D6 method to release ball with guide or support for spring /ball	
	D7 release the ball close to the vehicle	
	D8 detail on determining x e.g. difference between compressed length and original length	
	D9 method to ensure constant speed along track, e.g. friction compensate track / use of air track	
	D10 (relationship valid if a straight line produced) with (y-)intercept = $\frac{b}{kx^2}$	

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